

✓ Ch 9: Linear Regression

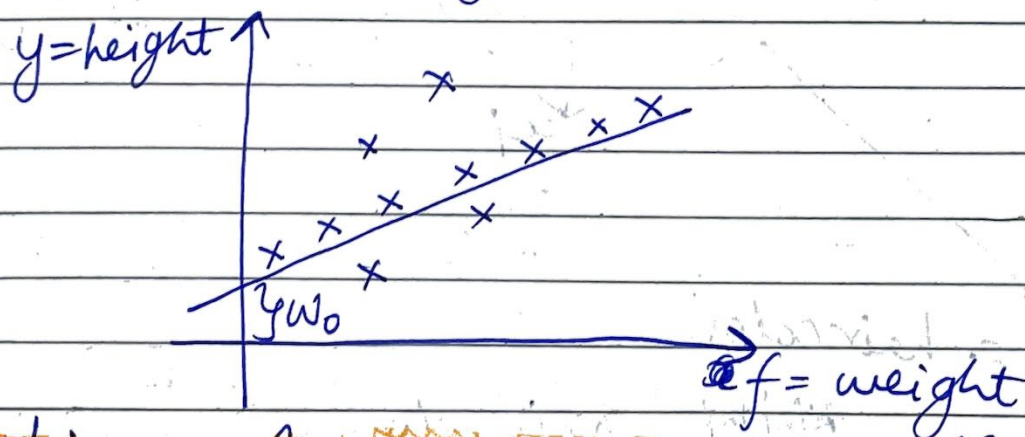
✓ ✓ Revision done

Ch 9 - Geometric Intuition of Linear Regression

It does actual regression unlike Logistic Regression (classification technique)

Ex - Predict height $\in \mathbb{R}$ given weight, gender, ethnicity, hair color, etc.

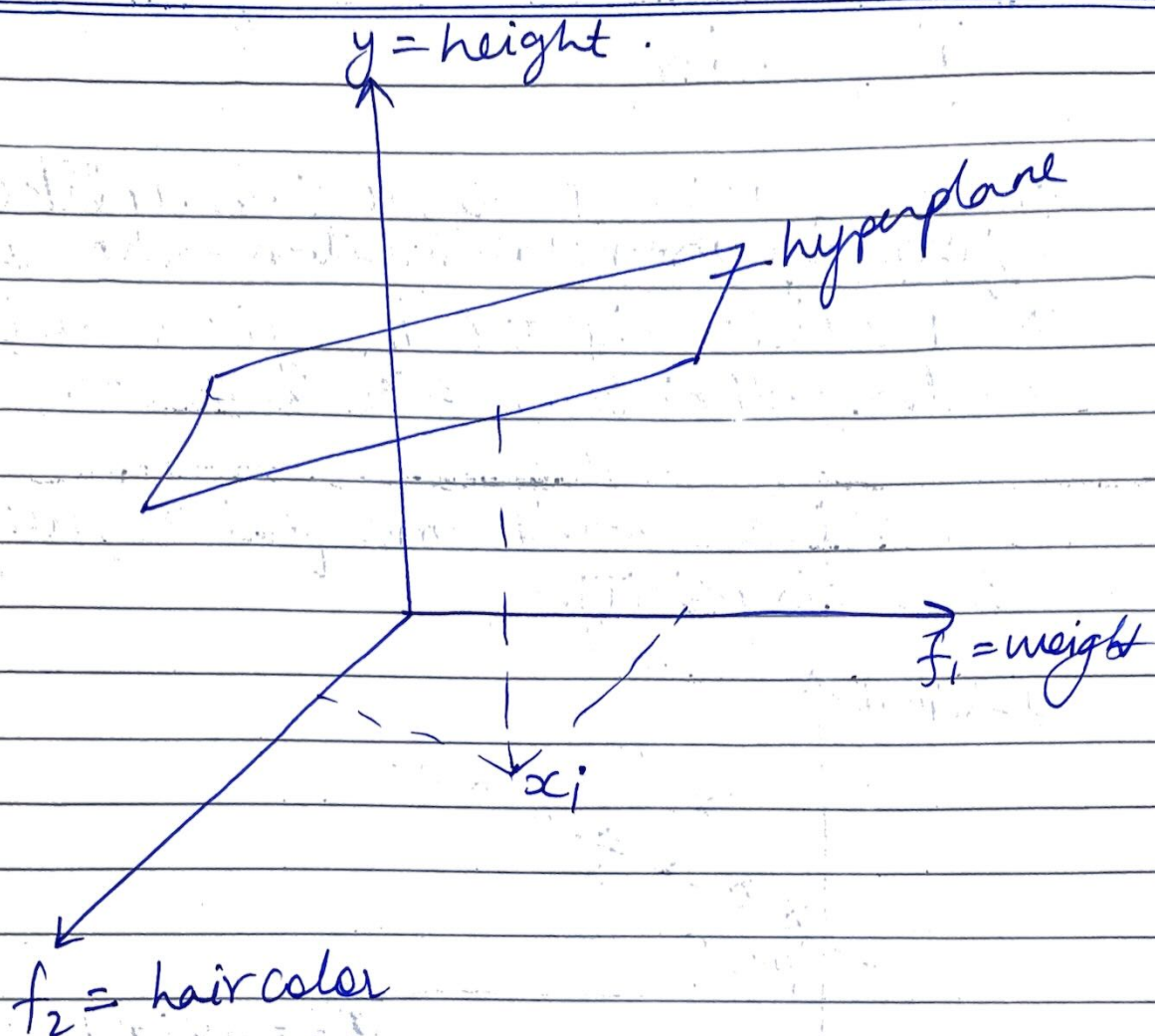
Consider only one feature, weight for simplicity.



Linear Regression $\rightarrow$ Find a line that best fits the given data.

$$\text{height} = \underbrace{w_1}_{\text{slope}} \times \text{weight} + \underbrace{w_0}_{\text{(y-intercept)}}$$
$$y = mx + c$$

In general, we have to predict a hyperplane that best fits the given data.



$$x_i = \langle \overset{\substack{\uparrow \text{weight}}}{x_{i1}}, \overset{\substack{\uparrow \text{hair color}}}{x_{i2}} \rangle \quad \text{In 3-D}$$

2 features

$$(\hat{y}_i) \text{ height} = W_1 * f_1 + W_2 * f_2 + W_0$$

= Equation of plane.

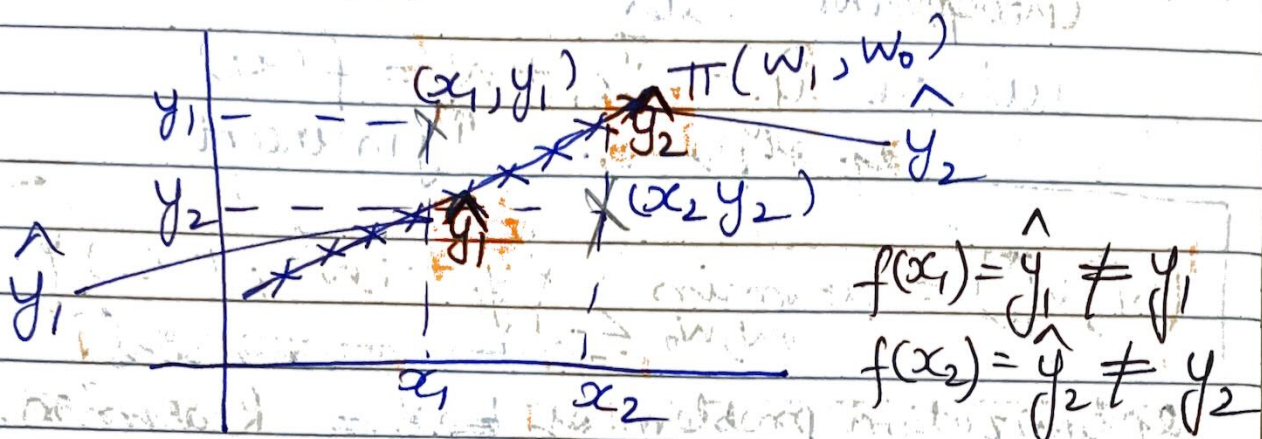
$$\text{So, } y_i = W^T x_i + W_0$$

All we are trying to find a plane that best fits the data.

Linear Regression means we are trying to find a linear plane or a line. ✓

Now, in 2-D:-

We say we have the model, $\pi(w_1, w_0)$ line



Let the prediction by model be denoted by $\hat{y}$.

So, for the model to best fit, the error must be minimum, i.e., difference b/w prediction and actual values.

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 \text{ --- Squared error.}$$

Actual - predicted

$$\text{error}_1 = y_1 - \hat{y}_1 = +ve$$

$$\text{error}_2 = y_2 - \hat{y}_2 = -ve$$

$$\text{error}_3 = 0$$

+ve/-ve depends whether point is above or below the best fit line

best fit line means = min sum of errors
2 types of errors, +ve and -ve. So we want (-ve) to be as close to 0 as possible

So we do square.

Optimization problem.

V2:

Mathematical Formulation

optimal

$$(W^*, W_0^*) = \underset{W, W_0}{\operatorname{argmin}} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Vector scalar Vector scalar squared error or loss

where $\hat{y}_i = W^T x_i + W_0 = f(x_i)$

if it doesn't go through origin $\nearrow$ y intercept

$$(W^*, W_0^*) = \underset{W, W_0}{\operatorname{argmin}} \sum_{i=1}^n \left\{ y_i - (W^T x_i + W_0) \right\}^2$$

square loss

Optimization problem of ~~Logistic~~ Linear Regression.

Linear Regression also called:

- 1) Ordinary Least Square
- 2) Linear Least Square

We also apply regularization:-

$$(W^*, W_0^*) = \underset{W, W_0}{\operatorname{argmin}} \sum_{i=1}^n \left\{ y_i - (W^T x_i + W_0) \right\}^2 + \lambda \|W\|_2^2$$

L2 norm λ controls balance b/w Sq. error and regularizer. L2 regularization.

Linear Regression with L2 reg = Ridge Regression
" " " L1 " = Lasso "
" " " L1+L2 = Elastic Net Regression.

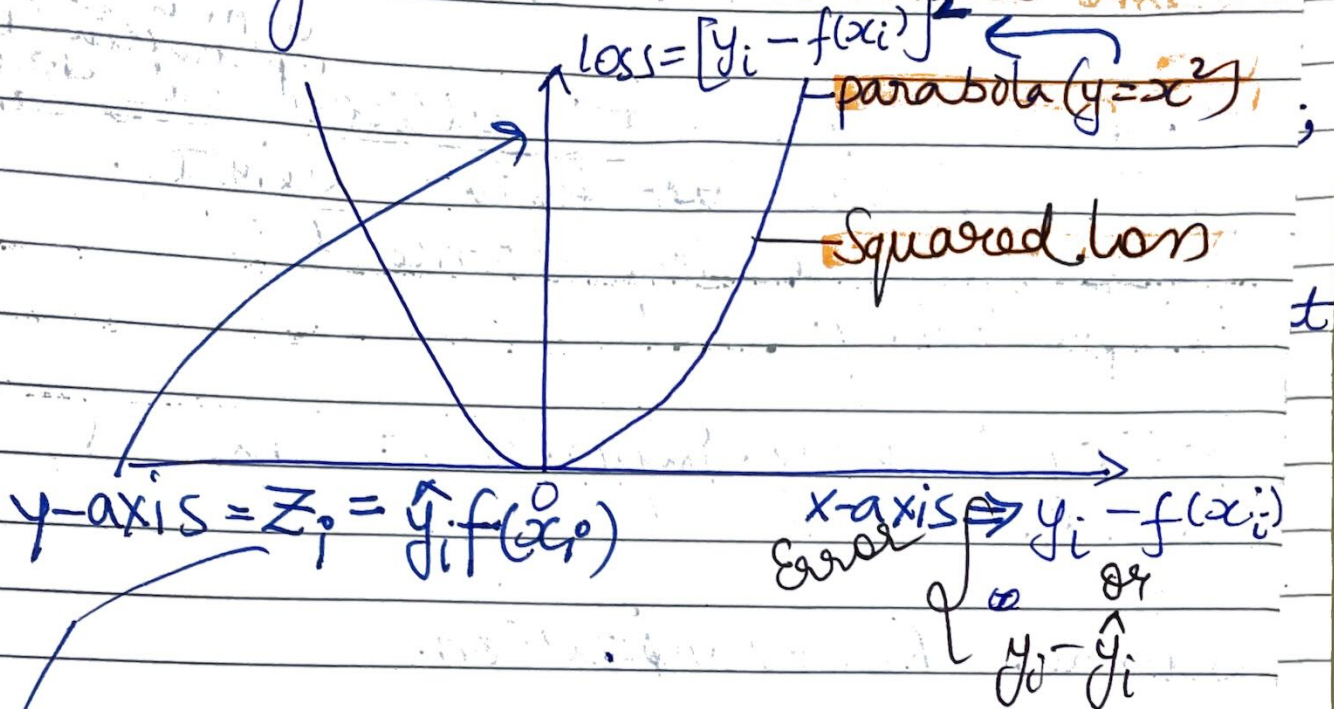
Regularizer pulls value of weights down when there is some noise in the data.

Andrew Ng - class at Stanford

geometric
probability
loss-minimization

Loss minimization Interpretation

Regression has error as x -axis is



So y axis is $(x\text{-axis})^2$.

Linear Regression has Squared loss
Note - 0-1 loss is used in classification.

In classification, if $Z_i = +ve$ - correct classification
 $Z_i = -ve$ - incorrect classification

In Regression (Linear) if error is either $(+ve)^2$ or $(-ve)^2$, loss is always +ve.

0-1 loss is for classification (binary)
squared loss is for regression (real value)

V3: Real World Cases

No problem of Imbalanced data in Lin Reg as there are no same as Logistic Regression. ^{are no classes}

If features are not multicollinear, ^{So no need of upsampling or down sampling} you can use ~~the~~ feature weights for feature Importance & Interpretability.

FE/FT is very important. Same as log. R.

Same Regularization.

Outliers :- (Outliers can impact Linear Regression Model)

In logistic regression, sigmoid function, limited the impact of outliers.

Nothing like that in Lin. Reg. So, we make use of filtering technique to filter out outliers.

- 1) Find hyperplane having outliers.
 - 2) Remove outliers from D_{train} .
 - 3) From new D_{train} , find new hyperplane again remove outliers from new D_{train} .
 - 4) Do it iteratively till you converge.
- This is called RANSAC (Random Sampling Consensus).

Why use regularization in Linear Regression.

As weights increase ^{to ∞} in Logistic Regression, regularization was introduced, but that problem doesn't occur in Linear Regression, then why have regularizer in Linear Regression at all.

Example: If we simulate a dataset and train the model to fit a function and find weights. But we also introduce some random error in real-world scenario. Weights in real world tend to increase. So, we introduce regularization so that weights don't go so high and stay within limits of just same as simulated example.